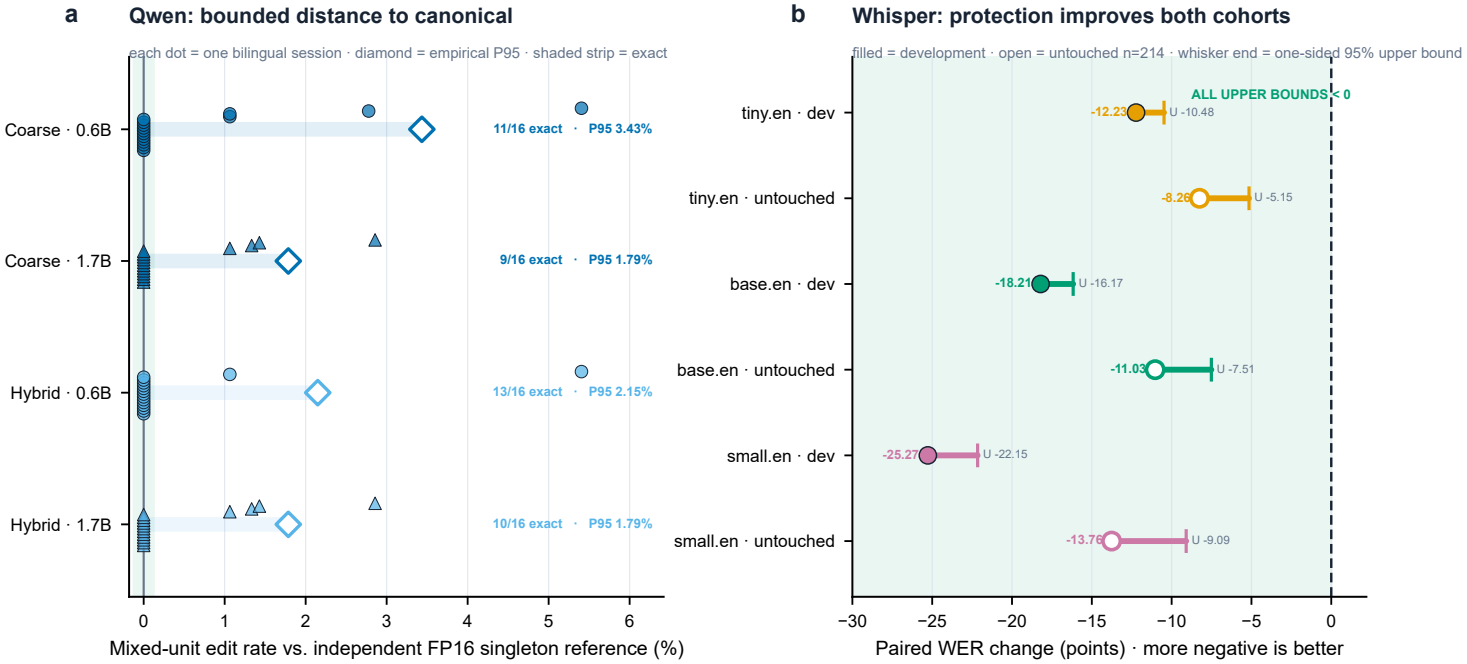

a
Qwen: bounded distance to canonical
each dot = one bilingual session · diamond = empirical P95 · shaded strip = exact
Coarse · 0.6B
Coarse · 1.7B
Hybrid · 0.6B
Hybrid · 1.7B
11/16 exact · P95 3.43%
9/16 exact · P95 1.79%
13/16 exact · P95 2.15%
10/16 exact · P95 1.79%
0
1
2
3
4
5
6
Mixed-unit edit rate vs. independent FP16 singleton reference (%)
b
Whisper: protection improves both cohorts
filled = development · open = untouched n=214 · whisker end = one-sided 95% upper bound
ALL UPPER BOUNDS < 0
tiny.en · dev
tiny.en · untouched
base.en · dev
base.en · untouched
small.en · dev
small.en · untouched
-12.23
U -10.48
-8.26
U -5.15
-18.21
U -16.17
-11.03
U -7.51
-25.27
U -22.15
-13.76
U -9.09
−30
−25
−20
−15
−10
−5
0
Paired WER change (points) · more negative is better